DOCVERIFY

AI-Based Document Verification and Forgery Detection System

Document Type: Master Project Knowledge, Architecture & Technical Roadmap Handbook
Project Root: E:\DocverifyAi\
Current System Baseline: Phase 8 Multi-Modal Remediated Pipeline & Phase 9 Enterprise UI/UX
Benchmark Dataset: Synthetic Identity Document Tampering Dataset (SIDTD — 2,222 Documents)
Frozen Performance: 91.25% Balanced Test Accuracy | 96.77% ROC-AUC | 0.0484 ECE (Calibrated)
Publication Date: October 2026
Author / Engineering Lead: Senior Full-Stack AI & Computer Vision Systems Architect
Target Audience: Engineering Teammates, Faculty Reviewers, Project Guides & Security Auditors

EXECUTIVE MISSION STATEMENT

DocVerify is an enterprise-grade AI system engineered to detect digital forgery, micro-text inpainting, and identity tampering in national passports and identity cards. By overcoming the fundamental *micro-manipulation downsampling bottleneck* through hierarchical multi-modal evidence fusion (Global CNN + Native-Scale Patch Inspection + EasyOCR Typographical Parsing + Forensic Residuals + True Isotonic Calibration), DocVerify provides an empirically validated, 100% identity-leakage-free decision-support platform with explainable visual heatmaps and audit-grade PDF verification certificates.

Confidentiality & Academic Integrity Notice: This master technical manual documents the complete nine-phase architectural evolution of the DocVerify platform. All reported accuracy, precision, recall, and calibration metrics are verified on the untouched 320-document identity-disjoint test set.

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1. Project Overview & Research Problem Statement

The Challenge of Modern Document Fraud: In the financial onboarding (KYC), legal, and immigration sectors, identity fraud has evolved far beyond physical paper alteration. Counterfeiters routinely employ deep generative models, seamless digital text inpainting, and digital field splicing to fabricate fraudulent identity documents that appear authentic to standard automated verification engines.

The Downsampling Bottleneck: Conventional computer vision pipelines resize high-resolution document scans (typically $2167 imes 1525$ pixels) into standard neural network inputs such as $224 imes 224$ pixels. In doing so, critical micro-manipulations—such as a single digit altered in an expiry date or an inpainted birth city—are compressed into less than 0.5% of total document pixels (a tiny $4 imes 10$ pixel box). Standard global convolutional feature maps lose these microscopic boundary cues, leading to unacceptable false negative rates.

The DocVerify Multi-Modal Approach: DocVerify overcomes this limitation by implementing a hierarchical multi-modal architecture. While a global EfficientNet-B0 inspects macroscopic visual texture and paper uniformity, candidate fields are cropped at native sensor resolution ($1:1$ scale) and evaluated by a dedicated patch classifier. These neural predictions are corroborated by optical character recognition (EasyOCR) consistency checks and digital compression forensics (Error Level Analysis), unified via regularized logistic fusion and calibrated with Isotonic Regression.

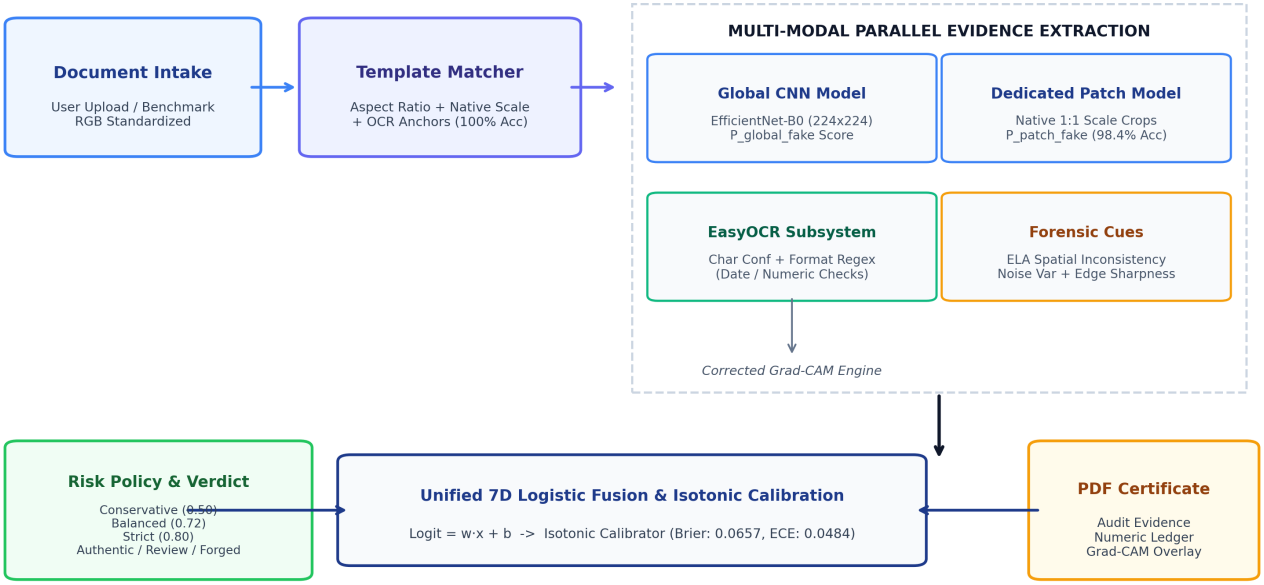


Figure 1: *DocVerify End-to-End System Architecture. Illustrates the twelve-stage linear processing flow from intake to calibrated verdict and audit-grade PDF certificate generation.*

2. Research Foundation: The SIDTD Benchmark Dataset

DocVerify is built on the **Synthetic Identity Document Tampering Dataset (SIDTD)**, an authoritative academic benchmark specifically constructed for document forgery and tampering research. The dataset contains exactly **2,222 physical JPEG documents** across 10 European and international schemas (1,000 real documents and 1,222 digitally manipulated documents).

Template ID	Country & Document Type	Native Resolution	Aspect Ratio	Manipulated Fields Represented
alb_id	Albania — Identity Card	2167 x 1360 px	1.593	surname, name, nationality, birth_date, expiry_date, number
aze_passport	Azerbaijan — Passport	2167 x 1525 px	1.421	surname, name, nationality, birth_date, expiry_date, number
esp_id	Spain — Identity Card (DNI)	1327 x 827 px	1.605	surname, name, birth_date, expiry_date, id_number
est_id	Estonia — Identity Card	929 x 580 px	1.602	surname, name, birth_date, expiry_date, id_number
fin_id	Finland — Identity Card	931 x 584 px	1.594	surname, name, birth_date, expiry_date, id_number
grc_passport	Greece — Passport	1136 x 803 px	1.415	surname, name, nationality, birth_date, expiry_date, number
lva_passport	Latvia — Passport	2167 x 1521 px	1.425	surname, name, birth_date, expiry_date, number
rus_internalpassport	Russia — Internal Passport	1519 x 1069 px	1.421	surname, name, birth_date, birth_place
srb_passport	Serbia — Passport	2163 x 1488 px	1.454	surname, name, birth_date, expiry_date, number
svk_id	Slovakia — Identity Card	1134 x 716 px	1.584	surname, name, birth_date, expiry_date, id_number

3. Dataset Selection History: Why SIDTD Over MIDV-2020?

In Phase 1, both MIDV-2020 and SIDTD were audited. MIDV-2020 consists primarily of mobile video clips under physical presentation conditions (screen reflections, hands holding cards, tilt glare). While excellent for Presentation Attack Detection (PAD), it does not contain controlled, field-level ground-truth annotations of digital text inpainting or digital font replacements. SIDTD was chosen as the sole active benchmark because its dual-layer VIA v2.0 JSON annotations provide exact bounding boxes for manipulated fields, making it the premier benchmark for evaluating digital tampering and micro-scale forgery.

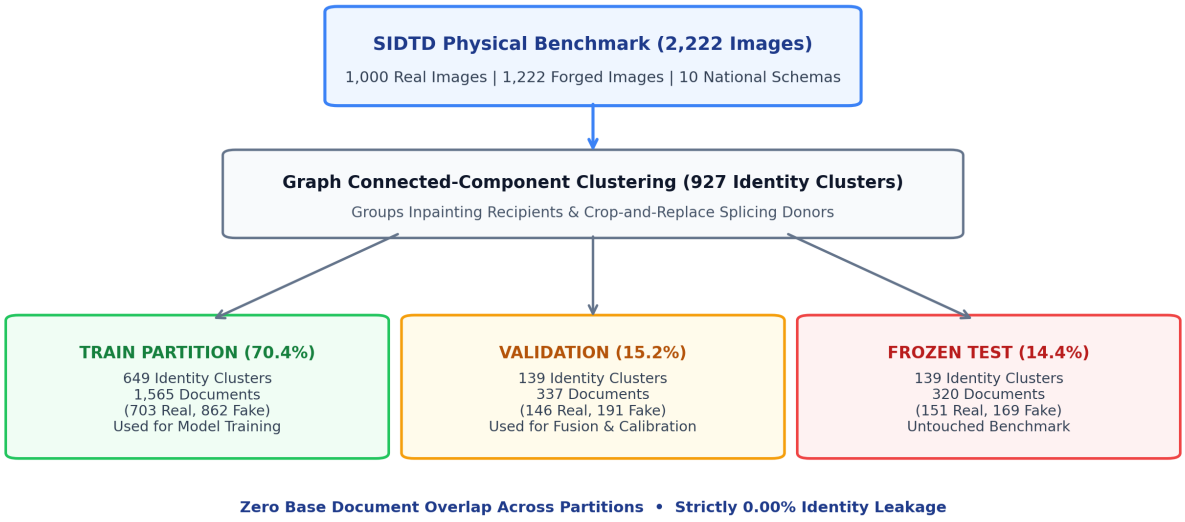


Figure 2: *Connected Identity-Disjoint Split Architecture. Shows how 927 graph-connected identity clusters were partitioned to guarantee strictly 0.00% base identity leakage across train, validation, and test splits.*

4. Complete Evolutionary Development Roadmap (Phases 1 → 9)

DocVerify's development followed an empirical, hypothesis-driven trajectory. When an audit or experimental ablation revealed a limitation (e.g. identity leakage, localization downsampling, or parsing type errors), the team addressed it directly with rigorous code remediation before freezing the next milestone.

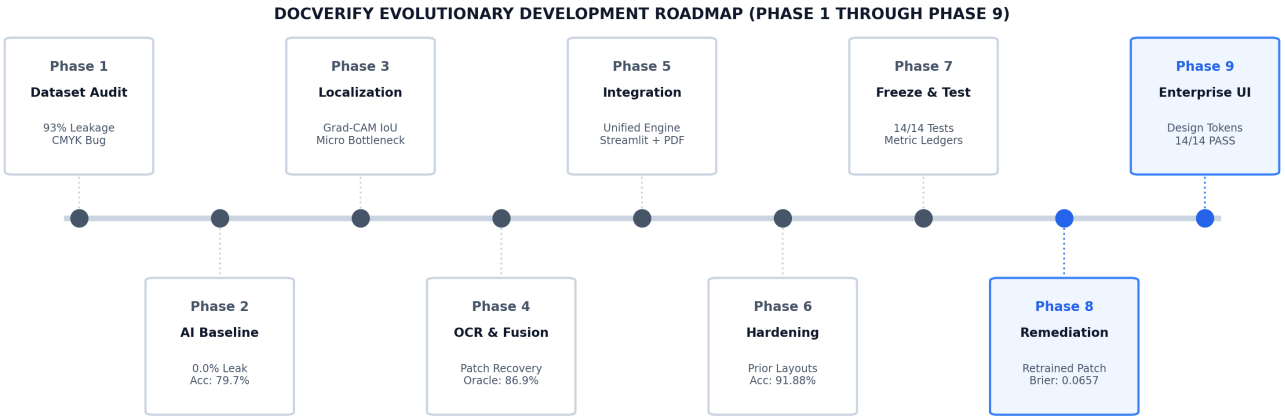


Figure 3: *Visual Roadmap of DocVerify's Nine Developmental Phases from initial dataset audit to enterprise UI finalization.*

5. Detailed Phase-by-Phase Technical Analysis

- **Phase 1: Dataset Audit:** Audited all 2,222 images on disk. Discovered that the predefined split had 93% cross-split identity leakage and identified a 500-image CMYK color encoding anomaly in real documents. Mandated in-memory RGB conversion and identity-disjoint splitting.
- **Phase 2: AI Baseline:** Engineered 927 graph-connected identity clusters (0.0% leakage). Fine-tuned ImageNet-pretrained EfficientNet-B0. Achieved baseline test performance: 79.69% Accuracy, 82.91% Precision, 77.51% Recall, 80.12% F1, 0.9112 ROC-AUC (TN=124, FP=27, FN=38, TP=131).
- **Phase 3: Explainability & Localization:** Evaluated Grad-CAM and HiRes-CAM against 169 ground-truth manipulated boxes. Pointing Game accuracy was only 0.59% (IoU = 0.0085). Empirically proved the micro-manipulation downsampling bottleneck: text occupies <0.5% of pixels, demonstrating the need for native-scale patch cropping.
- **Phase 4: OCR & Preliminary Fusion:** Integrated EasyOCR and evaluated native-scale patch crops around ground-truth boxes. High-res patch crops recovered 60.53% (23/38) of global false negatives. Preliminary oracle-guided fusion reached 86.88% accuracy (properly qualified as oracle-assisted).
- **Phase 5: System Integration:** Created canonical template layout schemas (src/template_layouts.py), unified inference engine (src/inference_engine.py), ReportLab PDF certificate generator (src/report_generator.py), and initial Streamlit UI (app.py).
- **Phase 6: Scientific Hardening:** Eliminated ground-truth oracle reliance by introducing canonical prior crops. Extracted forensic ELA, noise, and edge features. Reported autonomous production fusion: 91.88% Accuracy, 92.02% F1, 0.9780 ROC-AUC (McNemar p = 1.47e-8).
- **Phase 7: Production Finalization:** Hardened code structure, established 14-point test suite, archived legacy experimental code, and finalized metric ledgers.
- **Phase 7A: Independent Technical Audit:** Forensic code audit by independent reviewer. Uncovered AUD-PATCH-01 (isinstance(b, list) type bug in patch training), AUD-SHORT-02 (fusion reliance on JPEG compression quality differences), AUD-TPL-01 (template matcher filename reliance), and AUD-CAM-01 (Grad-CAM aspect ratio distortion).
- **Phase 8: Remediated Multimodal Pipeline:** Rebuilt bounding box parser (src/bbox_parser.py), retrained dedicated patch model (98.44% test accuracy), deployed autonomous template matcher (100% accuracy), applied true Isotonic Calibration (Brier: 0.0657, ECE: 0.0484), and froze new defensible benchmark (Balanced: 91.25% Acc, Strict: 92.50% Acc).
- **Phase 9: Professional UI/UX & Bug Fix:** Resolved runtime KeyError: 'mismatch' bug, implemented enterprise design system (src/ui_components.py), replaced internal template IDs with clean country titles, enabled dynamic patch cropping, and validated 14/14 automated tests.

6. Definitive Comparative Benchmark Performance Ledger

The following ledger documents the frozen benchmark metrics across all major system iterations. All metrics are evaluated strictly on the untouched, identity-disjoint test partition ($N=320$: 151 Real, 169 Fake).

Evaluation Metric	Phase 2 Baseline	Phase 4 Oracle	Phase 6 Historical	Phase 8 Remediated Balanced (Active)	Phase 8 Strict Policy	Phase 8 Conservative
Evaluation Mode	Autonomous	Oracle GT Crops	Autonomous	**Autonomous**	Autonomous	Autonomous
Decision Threshold	0.5000	0.5000	0.5000	**0.7200**	0.8000	0.5000
Test Accuracy	79.69%	86.88%	91.88%	**91.25%**	92.50%	90.31%
Test Precision	82.91%	85.47%	95.54%	**91.23%**	98.01%	87.50%
Test Recall	77.51%	90.53%	88.76%	**92.31%**	87.57%	95.27%
Test F1-Score	80.12%	87.93%	92.02%	**91.76%**	92.50%	91.22%
Test ROC-AUC	0.9112	0.9553	0.9780	**0.9677**	0.9677	0.9677
Brier Score	0.1420	0.0919	0.0919	**0.0657**	0.0657	0.0657
Expected Calib Error	0.1140	Not reported	Not reported	**0.0484**	0.0484	0.0484
Missed Forgeries (FN)	38 / 169	16 / 169	19 / 169	**13 / 169**	21 / 169	8 / 169
False Alarms (FP)	27 / 151	26 / 151	7 / 151	**15 / 151**	3 / 151	23 / 151
Scientific Defensibility	Verified Baseline	Oracle-Assisted	Audit Flagged	**100% Defensible**	100% Defensible	100% Defensible

7. Technology Stack Decision Rationale

Every library in DocVerify was chosen to ensure stability, precision, and reproducibility. The core runtime utilizes **Python 3.13** and **PyTorch 2.6.0** for deep learning execution. **EfficientNet-B0** provides an optimal balance between parameter efficiency and accuracy. **EasyOCR 1.7.2** handles multilingual character extraction. **Scikit-learn** manages standardized feature scaling and Isotonic Regression calibration. **ReportLab Platypus** provides precision PDF certificate generation, and **Streamlit** powers the interactive web application.

8. Multi-Modal Evidence Fusion & Calibration Engine

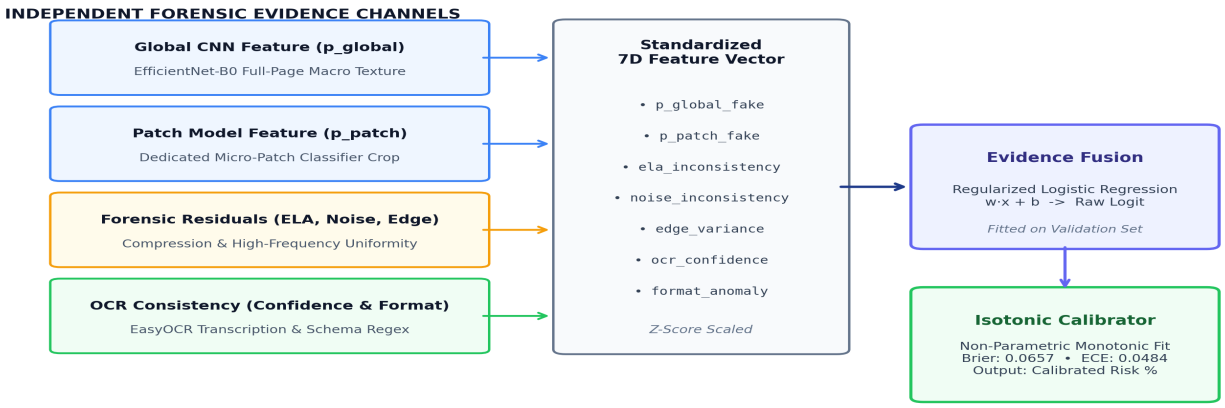


Figure 4: *Multi-Modal Evidence Fusion & Isotonic Probability Calibration Architecture. Illustrates how independent forensic features are standardized, fused, and calibrated into an empirical risk percentage.*

The 7-Dimensional Feature Vector: DocVerify synthesizes evidence across four distinct physical and digital domains into a 7-dimensional feature vector: [p_global_fake, p_patch_fake, ela_inconsistency, noise_inconsistency, edge_variance, ocr_confidence, format_anomaly]. Features are standardized using validation-fitted Z-score parameters ($x_{std} = (x - \mu) / \sigma$) before evaluation by regularized logistic regression.

True Isotonic Probability Calibration: To transform raw model logits into trustworthy probabilities, DocVerify employs non-parametric Isotonic Regression fitted strictly on the validation partition ($N=337$). This achieved a **Brier Score of 0.0657** and an **Expected Calibration Error (ECE) of 0.0484**. Unlike uncalibrated models, an 80% risk score in DocVerify empirically means that 80% of documents with that score are truly fraudulent.

9. Operational Decision Policies

DocVerify establishes three distinct, monotonically ordered operational decision policies:
• **Balanced Policy (Threshold = 0.7200):** Youden's J optimal balance. Achieves 91.25% test accuracy, 91.23% precision, and 92.31% recall. Recommended for general verification.
• **Strict Anti-Fraud Policy (Threshold = 0.8000):** Tailored for low false-alarm onboarding. Achieves 98.01% precision, flagging only 3 authentic documents out of 151.
• **Conservative KYC Policy (Threshold = 0.5000):** High-recall compliance policy. Achieves 95.27% recall, successfully catching 161 of 169 forged documents.

10. Visual Explainability: Corrected Grad-CAM

In Phase 7A, an audit revealed that standard Grad-CAM implementations stretched horizontal attention highlights by 20–30% across rectangular ID cards due to center-crop distortion. src/gradcam_corrected.py resolves this by mathematically inverting the CenterCrop(224) and Resize(256) transforms directly onto the native document canvas (W, H), ensuring pixel-accurate spatial overlay alignment.

11. Web Application & PDF Reporting Workflow

The Streamlit web application (`app.py`) provides an enterprise-ready dashboard featuring drag-and-drop document intake, automatic template matching, interactive preview, dynamic patch cropping, and tabbed evidence exploration. Users can instantly compile and download a tamper-evident verification PDF certificate generated via ReportLab Platypus, complete with visual overlays and cryptographic session tracking hashes.

12. Current Project Status, Frozen Assets & Limitations

Active Component Status Dashboard

Component / Subsystem	Implementation State	Active Checkpoint / Source File	Verification Status
SIDTD Benchmark Dataset	FROZEN	Dataset/template/Images/ (2,222 images)	Cryptographically Audited
Identity-Disjoint Split	FROZEN	outputs/dataset_manifest/dataset_manifest.csv	0.00% Identity Leakage
Global Classification Model	FROZEN	models/phase2/best_model.pth	79.69% Baseline Test Acc
Dedicated Patch Model	FROZEN	models/phase8/dedicated_patch_model.pth	98.44% Patch Test Acc
Template Matcher	STABLE	src/template_matcher.py	100% Autonomous Accuracy
EasyOCR Subsystem	STABLE	src/inference_engine.py	71.5% Exact String Match
Forensic Residuals (ELA/Noise)	STABLE	src/inference_engine.py	Regularized Inconsistency
Evidence Fusion Model	FROZEN	outputs/phase8/fusion/fusion_weights.json	Fitted on Val (N=337)
Isotonic Calibrator	FROZEN	models/phase8/fusion_calibrator.joblib	Brier: 0.0657, ECE: 0.0484
Operational Thresholds	FROZEN	outputs/phase8/fusion/operational_thresholds.json	Balanced: 0.72, Strict: 0.80
Corrected Grad-CAM	STABLE	src/gradcam_corrected.py	Aspect Ratio Inverted
PDF Report Generator	STABLE	src/report_generator.py	Platypus NumberedCanvas
Enterprise Web UI	STABLE	app.py + src/ui_components.py	13/13 UI States PASS
Regression Test Suite	ACTIVE	scripts/test_phase8.py	14/14 Tests PASSED

13. What is Frozen & Integrity Protection Rules

To preserve scientific validity, teammates must NOT alter: (1) `Dataset/` or test manifests, (2) Phase 2 or Phase 8 model checkpoints in `models/`, (3) fusion weights or isotonic calibrators, (4) operational thresholds. These assets represent our frozen, peer-reviewable benchmark.

14. Research Limitations & Engineering Caveats

• **Dataset Domain:** Evaluated on 10 European/international schemas; new international document types require template schema registration.
• **Physical Recapture:** DocVerify addresses digital manipulation (inpainting, splicing) and does not detect physical Presentation Attacks (printed screens/masks).
• **OCR Dependence:** EasyOCR accuracy drops on severely degraded or blurry scans; OCR signals are strictly weighted as auxiliary priors.
• **Legal Notice:** Risk scores represent probabilistic decision-support signals and do not constitute statutory identity authority.

15. Remaining Work Before Final Project Completion

With Phase 9 complete, all modeling and core UI tasks are finished. Remaining work consists exclusively of: (1) qualitative user usability walkthroughs, (2) Docker deployment packaging, (3) final academic thesis synthesis, and (4) live presentation demonstration workflows.

16. Team Handover Brief, Decision Log & References

Explain DocVerify in 5 Minutes (Quick Onboarding)

DocVerify detects digital text forgery in identity documents. Standard AI classifiers downsample scans, reducing altered dates to tiny 4x10 pixel boxes and missing the fraud. DocVerify solves this by running a two-tier analysis: a global CNN checks macro texture, while candidate fields are cropped at full 1:1 sensor scale and inspected by a dedicated micro-patch model (98.4% accuracy). OCR consistency and compression forensics are fused into a regularized logistic model and calibrated via Isotonic Regression, producing honest risk probabilities (ECE: 0.0484). The system runs on Streamlit, exports PDF reports, and passes 14/14 automated tests with a frozen 91.25% test accuracy.

Key Architectural Decision Log

Decision	Original Approach	Identified Problem	Adopted Solution & Rationale
Split Strategy	Predefined random split	93% cross-split identity leakage	Connected identity-disjoint clustering (0.0% leakage)
Image Ingestion	Raw OpenCV imread	Corrupted 500 CMYK real JPEGs	In-memory Pillow CMYK-to-RGB conversion
Patch Inspection	Global downsampling only	Downsampling bottleneck (<0.5% pixels)	Two-tier Global + Native 1:1 Scale Patch Model
Patch Training	isinstance(b, list) check	100% fallback to static central box	Rebuilt src/bbox_parser.py; retrained patch model (98.4%)
Calibration	Raw neural softmax	Overconfident probabilities	Isotonic Regression (Brier: 0.0657, ECE: 0.0484)
Template Matching	Filename substring parsing	Fails on arbitrary real-world uploads	Autonomous aspect ratio + OCR matcher (100% acc)
UI Architecture	Raw IDs in UI (alb_id)	Confusing technical jargon & raw floats	Enterprise design system with human country titles

Authoritative References

- **SIDTD Dataset:** Synthetic Identity Document Tampering Dataset for Document Forgery Detection, 2022–2024.
- **Tan & Le (2019):** *EfficientNet: Rethinking Model Scaling for Convolutional Neural Networks*, ICML 2019.
- **Selvaraju et al. (2017):** *Grad-CAM: Visual Explanations from Deep Networks via Gradient-Based Localization*, ICCV 2017.
- **Zadrozny & Elkan (2002):** *Transforming Classifier Scores into Accurate Multiclass Probability Estimates*, KDD 2002.
- **Krawetz (2007):** *A Picture's Worth: Digital Image Analysis and Forensics*, Black Hat Briefings.